Special topics in logic and security II *

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1. A code C consists of the following four words:

$$(0,0,0,0,0,0,0,0); (1,1,0,0,1,1,0,0); (0,0,1,1,0,0,1,1); (1,1,1,1,1,1,1,1).$$

   (a) Show that this code is linear over $\mathbb{F}_2$. Find its parameters $[n,k,d]$.
   (b) Find a generating matrix G for C.
   (c) Find a check-matrix H for C.
   (d) Show that $C \subset C^{\perp}$.
   (e) Find the minimal distance of $C^{\perp}$.

2. Let C be the following code over the field $\mathbb{F}_{11}$.

$$C = \{(f(1), f(2), \ldots, f(7)) \mid f(x) = ax^2 + bx + c \in \mathbb{F}_{11}[x]\}.$$

   (a) Show that one can always correct one mistake.
   (b) Correct the word $(6,3,4,9,7,9,8)$.

3. Let $S \subset \mathbb{C}$ be the set of complex numbers of radius $\frac{1}{\sqrt{2}}$. Let $B \subset \mathbb{C}^2$ be the set of all qubits. Show that $S \times S \subset B$. Show also that $B \neq S \times S$.

4. On a table at CERN there are two independent globes. The red globe has the quantum state $r = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$ while the blue globe has the quantum state $b = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$. We interpret by $|1\rangle$ that a mutation happens in the globe.

   (a) How probable is to separately observe mutations in these globes?
   (b) We consider the composite system of the two globes. Compute its quantum state.
   (c) How probable is to observe a mutation in the composite system?

5. Let H_n be a complex Hilbert space. A unitary function $U : H_n \otimes H_n \to H_n \otimes H_n$ such that for all states $|x\rangle, |y\rangle \in H_n$ one has:

$$U(|x\rangle|y\rangle) = |x\rangle|x\rangle$$

   is called a general quantum copy machine. Show that there is no general quantum copy machine.

6. OPTIONAL QUESTION

   Consider the triangles ABC with $A = 90°$ and $AB + BC = \frac{3}{2}$. Which of them has the maximal area?

*Master, 4-th semester.